For Each Simulation

0. Data: [#Text]

Text A character variable (up to 199 characters) that is printed when the file is read. The “#” character must be in column 1, and, accordingly, the variable starts in column 2. Any characters can be included in Text.

Note 1: Item 0 can be repeated multiple times.

Options must be proceeded with the "OPTIONS" specification and followed by the "END" specification. Letters can be specified as upper or lower case.

Optional character variables for UZF1 in any order, character variables must be proceeded by “OPTIONS” and followed by “END.” Total number of lines includes the number of character variables plus 1 line for the word “OPTIONS” and 1 line for the word “END.”:

0B. Data: [Options Block]

OPTIONS

[SPECIFYTHTR]

[SPECIFYTHTI]

[NOSURFLEAK]

[SPECIFYSURFK]

[REJECTSURFK]

[SEEPSURFK]

[ETSQUARE smoothfact]

[CAPILLARYUZET]

[NETFLUX unitrech unitdis]

[WCSAVE iwcun]

[WCREAD iwcrd]

END

SPECIFYSURFK -- An optional character variable. When SPECIFYSURFK is specified, the variable SURFK is specified in Data Set 4b.

REJECTSURFK -- An optional character variable. When REJECTSURFK is specified, variable SURFK instead of VKS is used to calculate rejected infiltration. REJECTSURFK is included only if SPECIFYSURFK is included.

SEEPSURFK -- An optional character variable. When SEEPSURFK is specified, variable SURFK instead of VKS is used to calculate surface leakage. SEEPSURFK is included only if SPECIFYSURFK is included.

ETSQUARE -- An optional character variable. When ETSQUARE is specified, groundwater ET is simulated using a constant potential ET rate, and is smoothed over a specified smoothing interval. This option is recommended only when using the NWT solver.

smoothfact -- An optional real variable specified if ETSQUARE is specified. For example, if the interval factor (smoothfact) is specified as smoothfact =0.1 (recommended value), then the smoothing interval will be calculated as: SMOOTHINT = 0.1*EXTDP and is applied over the range for groundwater head (h):

h < CELTOP-EXTDP, ET is zero;

CELTOP-EXTDP < h < CELTOP-EXTDP+SMOOTHINT, ET is smoothed; and

CELTOP-EXTDP+SMOOTHINT < h, ET is equal to potential ET.

CAPILLARYUZET – An optional character variable. When CAPILLARYUZET is specified, unsaturated zone ET is calculated using an alternate formulation (Niswonger, 2020). For this option, 3 additional variables are required in the stress period data for simulating ET, including AIR_ENTRY, H_ROOT, and ROOTACT. See input for stress periods below for details.

NETFLUX -- An optional character variable. When NETFLUX is specified, the sum of recharge (units of cubic length per time) and the sum of discharge (units of cubic length per time) is written to separate unformatted files using module UBDSV3.

Unitrech and Unitdis are the unit numbers to which these values are written when “SAVE BUDGET” is specified in Output Control. Values written to Unitrech are the sum of recharge values for the UZF, SFR2, and LAK Packages, and values written to Unitdis are the sum of discharge values for the UZF, SFR2, and LAK Packages. Values are averaged over the period between output times.

WCSAVE -- An optional character variable. When WCSAVE is specified, the UZF Package writes one volume-averaged water content for each model cell at the times when Output Control saves heads. iwcun is the unit number of the output file. List the file in the Name file as DATA(BINARY).

The average is taken above the water table. A cell that contains the water table is averaged only over the interval above the water table. A fully saturated cell is written as its saturated water content (THTS). Cells that are not active UZF cells are written as zero. One unformatted record is written for each layer that has SAVE HEAD set. The record uses the same layout as a binary head file, and the 16-character text label is UZF CELLAVG WC. A head-save unit must also be defined in Output Control. No record is written for a time step that does not save heads.

WCREAD -- An optional character variable. When WCREAD is specified, the initial water content is read from a file written by WCSAVE. iwcrd is the unit number of that file. List the file in the Name file as DATA(BINARY). WCSAVE and WCREAD must use different unit numbers.

The last time step stored in the file is used. Each layer record replaces the initial water content (THTI) for that layer. A value is kept inside the range from the residual water content to the saturated water

content. Cells written as zero are not changed. Item 7 is not read when WCREAD is specified, so a THTI array is not required even when the first stress period is transient. The water content is applied above the water table when the unsaturated-zone profile is built. A restart from these values should omit the steady-state stress period and should use the heads saved at the same time as the starting heads. A steady-state first stress period recalculates the water table before the transient solution.

1. Data: NUZTOP IUZFOPT IRUNFLG IETFLG IUZFCB1 IUZFCB2 NTRAIL2 NSETS2 NUZGAG SURFDEP

NUZTOP An integer value used to define which cell in a vertical column that recharge and discharge is simulated.

1 Recharge to and discharge from only the top model layer. This option assumes land surface is defined as top of layer 1.

2 Recharge to and discharge from the specified layer in variable IUZFBND. This option assumes land surface is defined as top of layer specified in IUZFBND.

3 Recharge to and discharge from the highest active cell in each vertical column. Land surface is determined as top of layer specified in IUZFBND. A constant head node intercepts any recharge and prevents deeper percolation. Appendix 1 29

4 Values of IUZFBND are used to define the top of the unsaturated zone, and recharge is added to the uppermost layer in a column that contains a water table. The layer to which recharge is added is updated at the beginning of each new time step and is held constant during a time step.

IUZFOPT An integer value equal to 0, 1, or 2. A value of zero indicates that flow will not be routed through the unsaturated zone, rather infiltration will be applied directly to the saturated zone. A value of 1 indicates that the vertical hydraulic conductivity will be specified within the UZF1 Package input file using array VKS. A value of 2 indicates that the vertical hydraulic conductivity will be specified within either the BCF, LPF or UPW Package input file. When IUZFOPT is greater than zero, unsaturated flow is routed by cell and the unsaturated-zone properties are read once for each model layer.

IRUNFLG An integer value that specifies whether ground water that discharges to land surface will be routed to stream segments or lakes as specified in the IRUNBND array (IRUNFLG not equal to zero) or if ground-water discharge is removed from the model simulation and accounted for in the ground-water budget as a loss of water (IRUNFLG=0). The Streamflow-Routing (SFR2) and(or) the Lake (LAK3) Packages must be active if IRUNFLG is not zero.

IETFLG An integer value that specifies whether or not evapotranspiration (ET) will be simulated. ET will not be simulated if IETFLG is zero, otherwise it will be simulated.

IUZFCB1 An integer value used as a flag for writing ground-water recharge, ET, and ground-water discharge to land surface rates to a separate unformatted file using subroutine UBUDSV. If IUZFCB1>0, it is the unit number to which the cell-by-cell rates will be written when “SAVE BUDGET” or a non-zero value for ICBCFL is specified in Output Control. If IUZFCB1 ≤ 0, cell-by-cell rates will not be written to a file.

IUZFCB2 An integer value used as a flag for writing ground-water recharge, ET, and ground-water discharge to land surface rates to a separate unformatted file using module UBDSV3. If IUZFCB2>0, it is the unit number to which cell-by-cell rates will be written when “SAVE BUDGET” or a non-zero value for ICBCFL is specified in Output Control. If IUZFCB2≤0, cell-by-cell rates will not be written to file.

NTRAIL2 An integer value equal to the number of trailing waves used to define the water-content profile following a decrease in the infiltration rate. The number of trailing waves varies depending on the problem, but a range between 10 and 20 is usually adequate. More trailing waves may decrease mass-balance error and will increase computational requirements and memory usage. This variable is not specified if IUZFOPT is zero or less.

NSETS2 An integer value equal to the number of wave sets used to simulate multiple infiltration periods. The number of wave sets should be set to 20 for most problems involving time varying infiltration. The total number of waves allowed within an unsaturated zone cell is equal to NTRAIL2*NSETS2. An error will occur if the number of waves in a cell exceeds this value. This variable is not specified if IUZFOPT is zero or less.

NUZGAG An integer value equal to the number of cells (one per vertical column) that will be specified for printing detailed information on the unsaturated zone water budget and water content. A gage also may be used to print the budget summed over all model cells.

SURFDEP A real value equal to the average undulation depth within a finite-difference cell.

Note 2: The variables NTRAIL2 and NSETS2 are used for dimensioning arrays.

2. Data: [IUZFBND (NCOL, NROW)] -- U2DINT

IUZFBND An array of integer values used to define the aerial extent of the active model in which recharge and discharge will be simulated. If IRUNFLG > 0: Read item 3

3. Data: [IRUNBND (NCOL, NROW)] -- U2DINT

IRUNBND An array of integer values used to define the stream segments within the Streamflow-Routing (SFR2) Package or lake numbers in the Lake (LAK3) Package that overland runoff from excess infiltration and ground-water discharge to land surface will be added. A positive integer value identifies the stream segment and a negative integer value identifies the lake number. 30 Documentation of the Unsaturated-Zone Flow Package

If IUZFOPT = 1: read item 4 once for each model layer.

4. Data: [VKS (NCOL, NROW)] -- U2DREL (one array for each model layer)

VKS An array of positive real values used to define the saturated vertical hydraulic conductivity of the unsaturated zone (LT-1). Note 3: Vertical hydraulic conductivity is optional when either the LPF, UPW, or BCF Package is used to define vertical hydraulic conductivity of cells (IUZFOPT=2).

If keyword SPECIFYSURFK is specified

4B. DATA [SURFK (NCOL, NROW)] -- U2DREL

SURFK An array of positive real values used to define the hydraulic conductivity (units of length per time). SURFK is used for calculating the rejected infiltration and/or surface leakage. If SURFK is set greater than VKS, then it is set equal to VKS.

5. Data: [EPS (NCOL, NROW)] -- U2DREL (one array for each model layer)

EPS An array of positive real values for each model cell used to define the Brooks-Corey epsilon of the unsaturated zone. Epsilon is used in the relation of water content to hydraulic conductivity (Brooks and Corey, 1966). This variable is not specified if IUZFOPT is zero or less.

If the character variable SPECIFYTHTR is specified in the options block, then items 5B is specified.

5B. Data: [THTR (NCOL, NROW)] -- U2DREL (one array for each model layer)

THTR An array of positive real values for each model cell used to define the Brooks-Corey residual water content of the unsaturated zone. THTR is used in the relation of water content to hydraulic conductivity (Brooks and Corey, 1966). This variable is not specified if IUZFOPT is zero or less.

6. Data: [THTS (NCOL, NROW)] -- U2DREL (one array for each model layer)

THTS An array of positive real values used to define the saturated water content of the unsaturated zone in units of volume of water to total volume (L3 L-3). This variable is not specified if IUZFOPT is zero or less.

7. Data: [THTI (NCOL, NROW)] -- U2DREL (one array for each model layer)

THTI An array of positive real values used to define the initial water content for each vertical column of cells in units of volume of water at start of simulation to total volume (L3 L-3). THTI should not be specified for simulations that start with a steady state stress period, unless the character variable SPECIFYTHTI is included in the OPTIONS block. This variable is not specified if IUZFOPT is zero or less. Item 7 is not read when WCREAD is specified.

When IUZFOPT is greater than zero, unsaturated flow is routed through each active model cell above the water table. Items 4, 5, 5B, 6, and 7 are read once for each model layer, beginning with layer 1 and continuing through layer NLAY. The array entered for a layer is the property of that cell. Item 4 is repeated for each layer only when IUZFOPT equals 1. When IUZFOPT equals 2, vertical hydraulic conductivity is taken from the BCF, LPF, or UPW Package for each layer, and items 5, 6, and, when they are required, items 5B and 7 are still read for each layer. Item 4B (SURFK) remains a single array for the land surface. A model with one layer uses one array for each of these items. Item 7 is read for each layer when the first stress period is transient or when SPECIFYTHTI is specified, unless WCREAD is specified.

If NUZGAG>0: Item 8 is repeated NUZGAG times

8. Data: [IUZROW] [IUZCOL] IFTUNIT [IUZOPT]

IUZROW An integer value equal to the row number of the cell that unsaturated-zone information will be printed for each time step. The variable is not used when IFTUNIT is negative.

IUZCOL An integer value equal to the column number of the cell that unsaturated-zone information will be printed for each time step. The variable is not used when IFTUNIT is negative.

IFTUNIT An integer value equal to the unit number of the output file. A positive value is for output of individual cells whereas a negative value is for output that is summed over all model cells. The summed output includes applied infiltration, runoff, actual infiltration, ground-water discharge to land surface, ET from the unsaturated zone, ET from ground water, recharge, and change in unsaturated-zone storage.

IUZOPT An integer value that is a flag for the type of expanded listing desired in the output file. The variable is not used when IFTUNIT is negative.

1 Prints time, ground-water head, and thickness of unsaturated zone, and cumulative volumes of infiltration, recharge, storage, change in storage and ground-water discharge to land surface.

2 Same as option 1 except rates of infiltration, recharge, change in storage, and ground-water discharge also are printed.

3 Prints time, ground-water head, thickness of unsaturated zone, followed by a series of depths and water contents in the unsaturated zone. This option cannot be used when IUZFOPT is zero or less.

Note 4: Item 8 must include exactly NUZGAG lines (records) of data. If NUZGAG > 1, it is permissible to repeat the same cell if a summary of volumes and rates and water contents are desired for a given cell. The records within item 8 can be listed in any order.

Note 5: A unique unit number for IFTUNIT must be specified for each line or record in item 8 and each number must match that used in the MODFLOW name file as Ftype “DATA” file types and file names (see Harbaugh, 2005).

For Each Stress Period

9. Data: NUZF1

NUZF1 An integer value for reusing or reading infiltration rates that can change each stress period. If NUZF1 > 0, then infiltration rates at land surface are specified. If NUZF1 < 0, then infiltration rates from the previous stress period are used.

If NUZF1 > 0:

10. Data: [FINF (NCOL, NROW)] – U2DREL

FINF An array of positive real values used to define the infiltration rates (LT-1) at land surface for each vertical column of cells.

Note 6: If FINF is specified as being greater than the vertical hydraulic conductivity then FINF is set equal to the vertical unsaturated hydraulic conductivity. Excess water is routed to streams or lakes when IRUNFLG is not zero, and if SFR2 or LAK3 is active.

If IETFLG > 0: Read Items 11 through 16

11. Data: NUZF2

NUZF2 An integer value for reusing or reading ET demand rates that can change each stress period. If NUZF2>0, then ET demand rates are specified.

If NUZF2>0:

12. Data: [PET (NCOL, NROW)] – U2DREL

PET An array of positive real values used to define the ET demand rates (L1 T-1) within the ET extinction depth interval for each vertical column of cells.

13. Data: NUZF3

NUZF3 An integer value for reusing or reading ET extinction depths that can change each stress period. If NUZF3>0, then ET extinction depths are specified.

If NUZF3>0:

14. Data: [EXTDP (NCOL, NROW)] – U2DREL

EXTDP An array of positive real values used to define the ET extinction depths.

Note 7: The quantity of ET removed from a cell is limited by the volume of water stored in the unsaturated zone above the extinction depth. If ground water is within the ET extinction depth, then the rate removed is based on a linear decrease in the maximum rate at land surface and zero at the ET extinction depth. The linear decrease is the same method used in the Evapotranspiration Package (McDonald and Harbaugh, 1988, chap. 10).

15. Data: NUZF4

NUZF4 An integer value for reusing or reading the extinction water content that can change each stress period. If NUZF4>0, then extinction water contents are specified.

If NUZF4>0:

16. Data: [EXTWC (NCOL, NROW)] – U2DREL

EXTWC An array of positive real values used to define the extinction water content below which ET cannot be removed from the unsaturated zone. This variable is not specified if IUZFOPT is zero or less. Note 8: EXTWC must have a value between (THTS-Sy) and THTS, where Sy is the specific yield specified in either the LPF or BCF Package.

If the character variable CAPILLARYUZET is specified in the options block, then items 17-22 are specified.

17. Data: [NUZF5]

NUZF5 An integer value for reusing or reading the air entry pressure that can change each stress period.

If NUZF5>0, then air entry pressure values are specified.

18. Data: [AIR_ENTRY (NCOL, NROW)] – U2DREL

AIR_ENTRY An array of positive real values used to define the Brooks-Corey air-entry pressure variable for calculating capillary pressure that is used in the unsaturated zone.

19. Data: [NUZF6]

NUZF6 An integer value for reusing or reading the root pressure that can change each stress period.

If NUZF6>0, then root pressure values are specified.

20. Data: [H_ROOT (NCOL, NROW)] – U2DREL

H_ROOT An array of positive real values that are the root pressures used for simulating ET in the unsaturated zone.

21. Data: [NUZF7]

NUZF7 An integer value for reusing or reading the root pressure that can change each stress period.

If NUZF7>0, then root activity values are specified.

22. Data: [ROOTACT (NCOL, NROW)] – U2DREL

ROOTACT An array of positive real values that are the root activity values used for simulating ET in the unsaturated zone.